Lab 3.2 source pack

All records are fictional training data. They describe no actual facility, person or organisation. Values are teaching assumptions, not approved design criteria. Do not use outputs to operate, maintain or approve real equipment.

process-data.csv

sample_id: 1; fill_L: 9.9; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 2; fill_L: 10.1; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 3; fill_L: 10.0; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 4; fill_L: 10.2; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 5; fill_L: 9.8; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 6; fill_L: 10.1; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 7; fill_L: 9.9; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 8; fill_L: 10.0; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 9; fill_L: 10.2; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 10; fill_L: 10.1; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 11; fill_L: 9.9; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 12; fill_L: 9.8; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 13; fill_L: 10.0; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 14; fill_L: 10.1; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 15; fill_L: 10.2; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 16; fill_L: 9.9; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 17; fill_L: 10.0; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 18; fill_L: 10.1; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 19; fill_L: 9.8; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 20; fill_L: 10.0; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 21; fill_L: 10.2; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 22; fill_L: 10.1; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 23; fill_L: 9.9; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 24; fill_L: 10.0; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 25; fill_L: 10.1; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 26; fill_L: 9.8; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 27; fill_L: 10.0; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 28; fill_L: 10.2; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 29; fill_L: 9.9; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 30; fill_L: 10.1; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 31; fill_L: 10.0; phase: monitoring; LSL_L: 9.4; USL_L: 10.6

sample_id: 32; fill_L: 10.2; phase: monitoring; LSL_L: 9.4; USL_L: 10.6

sample_id: 33; fill_L: 10.8; phase: monitoring; LSL_L: 9.4; USL_L: 10.6

sample_id: 34; fill_L: 10.1; phase: monitoring; LSL_L: 9.4; USL_L: 10.6

sample_id: 35; fill_L: 9.9; phase: monitoring; LSL_L: 9.4; USL_L: 10.6

defects.csv

category: Calibration evidence; count: 12

category: Blocked strainer; count: 8

category: Air ingress; count: 5

category: Other; count: 5

dmaic-template.md

DMAIC improvement report
Define
Problem, CTQ, scope and stakeholder
Measure
Dataset version, units, missing values, baseline and limitations
Analyse
Pareto, causal hypotheses, 5-Whys, chart, limits and capability caveats
Improve
Proposed trial, owner, predicted effect and approval gate
Control
Metric, review frequency, reaction plan and owner
Sources and sign-off